CE-03: MicroMacro Error vs. Beta Ratio

Experiment Report — Two-Scale CTMC Project

Claude Experiments Pipeline

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Abstract

This report quantifies how the ratio $r = \beta_{\text{out}}/\beta_{\text{in}}$ between inter- and intra-community infection rates affects the MicroMacro approximation error. Five scenarios sweep $r \in \{0.1, 0.3, 1.0, 2.0, 5.0\}$ on a fixed chain-of-cliques topology ($L = 10$, $n = 100$, $d = 6$, $k = 5$) with 100 paired replications each. The primary finding is a **sign reversal** in ΔRMST: MicroMacro arrives *later* than full Gillespie at low r ($\Delta\text{RMST} = +4796$ at $r = 0.1$), and *earlier* at high r ($\Delta\text{RMST} = -1355$ at $r = 1.0$, -1638 at $r = 2.0$). The crossover occurs between $r = 0.3$ and $r = 1.0$. At $r \geq 2$ the full-Micro epidemic rarely reaches the final community within the simulation horizon (completion 54% and 1% respectively), while MicroMacro completes in all runs—indicating a qualitative false-positive bias under strong inter-community coupling. *Ukrainian summary*: see `docs/ce03_beta_ratio/report.md`.

1 Experimental Design

1.1 Objective

Determine how the ratio of inter-community to intra-community infection rate modulates MicroMacro approximation error, and characterise the direction (premature vs. delayed arrival) across the parameter range.

1.2 Network and Parameters

- Topology: chain-of-cliques, $L = 10$ communities, $n = 100$ nodes each, degree $d = 6$, $k = 5$ inter-community edges per boundary.
- Intra-community infection rate: $\beta_{\text{in}} = 0.003$.
- Inter-community infection rate: $\beta_{\text{out}} = r \cdot \beta_{\text{in}}$.
- Rate ratios: $r \in \{0.1,\ 0.3,\ 1.0,\ 2.0,\ 5.0\}$.
- Recovery: $\gamma = 0$ (pure SI model).
- Replications: 100 paired runs per scenario (same network and seed).
- Horizon T_{end}: scenario-dependent (20,000 to 2,000; shorter for larger r to keep runtimes feasible).

1.3 Primary Metric

$$\Delta\text{RMST} = \overline{T_{\text{MM}}} - \overline{T_{\text{Micro}}}$$

where arrival times use restricted-mean estimates with censoring at T_{end} (censored runs contribute T_{end} to the mean). A positive ΔRMST indicates MicroMacro predicts *delayed* arrival; negative indicates *premature* arrival. Bootstrap 95% confidence intervals use 10,000 resamples. Relative

Wasserstein distance rel-W_1 measures distributional discrepancy normalised by the Micro mean arrival time.

2 Results

Table 1: CE-03 results: MicroMacro error by beta ratio. ΔRMST = censored-mean difference (MicroMacro − Micro). $r = \beta_{\text{out}}/\beta_{\text{in}}$. "Comp." = fraction of Micro runs reaching the final community by T_{end} (MicroMacro: 1.00 for all scenarios). All CIs exclude zero.

r	β_{out}	ΔRMST	95% CI		rel-W_1	Micro comp.
			lower	upper		
0.1	0.0003	+4796	+4384	+5213	1.624	1.00
0.3	0.0009	+562	+300	+812	0.297	1.00
1.0	0.0030	−1355	−1494	−1220	0.456	1.00
2.0	0.0060	−1638	−1746	−1527	0.558	0.54
5.0	0.0150	−1213	−1283	−1141	0.566	0.01

2.1 Key Observations

1. **Sign reversal.** ΔRMST is positive at $r \leq 0.3$ and negative at $r \geq 1.0$. The crossover r^* lies between 0.3 and 1.0.

2. **Large magnitude at extremes.** $|\Delta\text{RMST}| \approx 4800$ at $r = 0.1$ and ≈ 1600 at $r = 2.0$—both large relative to the horizon.

3. **Distributional divergence.** rel-$W_1 = 1.624$ at $r = 0.1$ (distributions differ by more than the Micro mean arrival time).

4. **Completion collapse at high r.** At $r = 2.0$ only 54% of full-Micro runs reach the final community within $T_{\text{end}} = 3000$; at $r = 5.0$ only 1%. MicroMacro always reports full traversal, yielding a 99 percentage-point false-positive epidemic rate at $r = 5.0$.

3 Interpretation

3.1 Mechanism at Low r (delayed arrival, ΔRMST > 0)

When $\beta_{\text{out}} \ll \beta_{\text{in}}$, intra-community spread is fast and inter-community spread is rare. In the full Gillespie (Micro) model, the $k = 5$ gateway nodes on the boundary of the source community can fire their cross-community edges as soon as *they themselves* become infected, regardless of the community-wide infected fraction $\bar{i}(t)$.

MicroMacro uses the aggregate import hazard $\lambda_{\text{macro}}(t) = \beta_{\text{out}} \cdot \bar{i}(t)$, proportional to the community-wide mean. When β_{out} is tiny, this hazard remains negligible until $\bar{i}(t)$ grows substantially, so the macro import event fires *late*. The direct gateway edges in Micro can fire earlier, giving ΔRMST > 0.

3.2 Mechanism at High r (premature arrival, ΔRMST < 0)

When $\beta_{\text{out}} \geq \beta_{\text{in}}$, the aggregate hazard $\lambda_{\text{macro}}(t) = \beta_{\text{out}} \cdot \bar{i}(t)$ becomes large as soon as any infecteds appear in the source community. MicroMacro fires imports aggressively and places each at a uniformly random node in the target community, artificially accelerating chain traversal.

In full Micro, the epidemic must propagate through only $k = 5$ physical edges per boundary. Even with large β_{out}, traversing nine consecutive boundaries in a chain is constrained by within-community buildup at each step. Micro rarely completes the chain within a short T_{end} (1% at $r = 5.0$), while MicroMacro always does.

3.3 Comparison with CE-01 Baseline

CE-01 used homogeneous rates ($r = 1$) and found $\Delta\text{RMST} = -209$ with rate abstraction as the dominant error source. CE-03 at $r = 1.0$ gives $\Delta\text{RMST} = -1355$; the larger magnitude is consistent with a shorter $T_{\text{end}} = 4000$ (vs. 5000 in CE-01). Both confirm premature arrival at $r = 1$.

4 Conclusion

CE-03 reveals a qualitative change in MicroMacro error direction as a function of r:

- **Low r ($r < r^* \approx 0.5$–0.8):** MicroMacro predicts *delayed* arrival. The aggregate hazard underweights early gateway activity, causing the macro import to fire too late.
- **High r ($r > r^*$):** MicroMacro predicts *premature* arrival. The aggregate hazard overestimates the effective multi-boundary crossing rate. At $r = 5.0$ the qualitative prediction is wrong: Micro rarely spreads across the full chain, while MicroMacro always does (99% false-positive rate).
- **Crossover near $r^* \approx 0.5$–0.8:** Both error directions cancel. This cancellation is topology-dependent, not a regime of correct approximation.

A single scalar correction to β_{out} cannot remove the bias across the full parameter range. Structural corrections—such as using the actual gateway-node infected count rather than the community mean—are needed.

Data Location

`output/claude_experiments/ce03_beta_ratio/summary.csv`

Related Experiments

CE-01 (ablation: rate abstraction vs. placement), CE-04 (SIR: recovery effect).